图论基础练习

人员

于家瑞、陶汇笙、于霄龙、王恩泽、郭栩睿、洪晨栋、洪晨棋 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3296

比赛概况 题目列表 选择题列表 提交记录 ★实时榜单

王向东老师周六十点半C++kruskal

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E
1	yujiarui	于家瑞	500	11023	100	100	100	100	100
2	guoxurui	郭栩睿	400	8721	100	100	100	100	
3	yuxiaolong	于霄龙	300	7711		100	100	100	
4	wangenze	王恩泽	300	7758		100	100	100	
5	taohuisheng	陶汇笙	300	8753	100	100	100		
6	cuichenhe	崔宸赫	200	8329	100	100			

本周作业

https://cppoj.kids123code.com/contest/3405 (课上讲了 A ~ C 这些题, 课后作业是 C 题)

课堂表现

同学们课上整体做题还不错, 有几位同学最小生成树不熟的需要课下复习。

课堂内容

Shortest Path 3

dijkstra 类模板题, 跟模板唯一的区别是这道题既有点权又有边权, 在松弛的时候考虑上点权即可

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2e5 + 5;
struct node {
  int dis, id;
  bool operator < (const node& p) const { return dis < p.dis; }
  bool operator > (const node& p) const { return dis > p.dis; }
};
vector<node> vec[maxn];
int w[maxn], f[maxn];
```

```
bool st[maxn];

signed main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  for (int i = 1; i <= m; ++i) {
    int a, b, c; cin >> a >> b >> c;
    vec[a].push_back({c,b}), vec[b].push_back({c,a});
  }

  memset(f, 0x3f, sizeof(f));
  priority_queue<node, vector<node>, greater<node>> q;
  q.push({w[1],1}); f[1] = 0;
  while (!q.empty()) {
    node u = q.top(); q.pop();
    int dis = u.dis, id = u.id;
    if (st[id]) continue;

    st[id] = true;
    for (node it : vec[id]) {
      if (st[it.id]) continue;
      if (dis+it.dis+w[it.id] < f[it.id]) {
        f[it.id] = dis+it.dis+w[it.id]; q.push({f[it.id],it.id});
      }
    }
  }

  for (int i = 2; i <= n; ++i) cout << f[i] << " ";
  cout << endl;
  return 0;
}
```

营救

最小生成树, 直到把 s 和 t 连到同一个并查集中为止

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e4 + 5;
struct node {
  int u, v, value;
} w[maxn];

bool cmp(node p, node q) {
  return p.value < q.value;
}

int f[maxn];
```

```
int fFind(int x) {
  if (f[x] != x) f[x] = fFind(f[x]);
  return f[x];
}

int main()
{
  int n, m, s, t; cin >> n >> m >> s >> t;
  for (int i = 1; i <= n; ++i) f[i] = i;

  for (int i = 1; i <= m; ++i) cin >> w[i].u >> w[i].v >> w[i].value;
  sort(w+1, w+m+1, cmp);

  for (int i = 1; i <= m; ++i) {
    int u = w[i].u, v = w[i].v, value = w[i].value;
    int fu = fFind(u), fv = fFind(v);
    if (fu != fv) f[fu] = fv;

    if (fFind(s) == fFind(t)) { cout << value << endl; break; }
  }
  return 0;
}
```

通往奥格瑞玛的道路

二分 + dijkstra

二分城市收费的最大值, 后续 check 时只能选收费更低的城市, 求起点到终点的最短路, 看最短路是否符合题目要求

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e4 + 5;
const int inf = 0x3f3f3f3f3f3f3f3f;
struct edge {
    int to, value;
};
vector<edge> vec[maxn];

struct node {
    int d, id;
    bool operator < (const node& p) const { return d < p.d; }
    bool operator > (const node& p) const { return d > p.d; }
};
int dis[maxn], st[maxn];

int w[maxn];
```

```cpp
bool check(int mid, int n, int B) {
    if (w[1]>mid || w[n]>mid) return false;

    memset(dis, 0x3f, sizeof(dis)), memset(st, false, sizeof(st));
    priority_queue<node, vector<node>, greater<node>>q;
    dis[1] = 0, q.push({dis[1],1});

    while (!q.empty()) {
        node u = q.top(); q.pop();
        int d = u.d, id = u.id;
        if (st[id]) continue;
        st[id] = true;

        for (edge it : vec[id]) {
            if (w[it.to] > mid) continue;
            if (d+it.value < dis[it.to]) {
                dis[it.to] = d+it.value, q.push({dis[it.to],it.to});
            }
        }
    }

    return dis[n]<=B;
}

signed main()
{
    int n, m, B; cin >> n >> m >> B;
    for (int i = 1; i <= n; ++i) cin >> w[i];
    while (m -- ) {
        int a, b, c; cin >> a >> b >> c;
        vec[a].push_back({b,c}), vec[b].push_back({a,c});
    }

    int l = 0, r = 1e9;
    while (l <= r) {
        int mid = (l + r) / 2;
        if (check(mid, n, B)) r = mid-1;
        else l = mid+1;
    }

    if (l > 1e9) cout << "AFK" << endl;
    else cout << l << endl;
    return 0;
}
```